



C23-M/MRAC-402

23446

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – FOURTH SEMESTER EXAMINATION

HYDRAULICS AND FLUID POWER SYSTEMS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define the following properties of fluid :
 - (a) Viscosity
 - (b) Specific gravity
2. Calculate the density of an oil whose specific gravity is 0·8.
3. Define the following terms related to the fluid flow :
 - (a) Steady flow
 - (b) Unsteady flow
4. State Bernoulli's theorem.
5. Find the loss of head due to friction in a pipe of 1 m diameter and 10 km long. The velocity of water in the pipe is 1·2 m/sec. Take friction factor, $f = 0·002$.

6. Define the term Total Energy Line (TEL).
7. Classify water turbines according to action and direction of water over runner.
8. Define the term 'Cavitation' referring to the pumps.
9. List any three applications of the fluid power systems.
10. List any three actuators used in the fluid power systems.

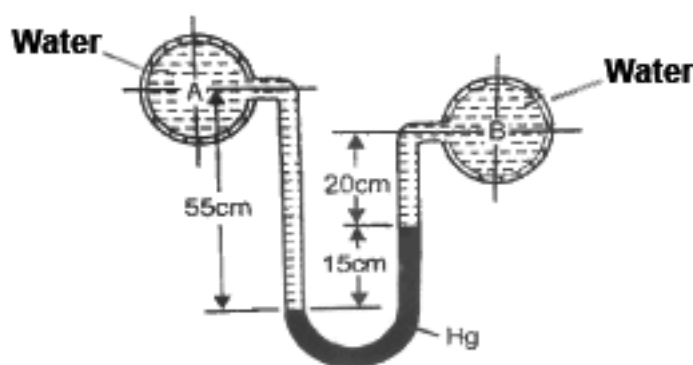
PART—B

10×5=50

- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. A differential mercury manometer connected at two points in a pipe contains water as shown in the figure below. Determine the pressure difference between the two points in kPa.

10



12. Oil flows through a horizontal venturimeter 20 cm inlet and 15 cm throat diameter. If the absolute pressure at inlet is 150 kPa and throat is 125 kPa, find the discharge of oil through the meter. Assume coefficient of meter as 0.97 and specific gravity of oil as 0.8.

10

/23446

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[Contd...

- 13.** Water is flowing through a tapered pipe having diameters 300 mm at section (1) and 200 mm at section (2). The discharge through the pipe is 180 litres/sec. The section (1) is 8 m above the datum and section (2) is 4 m above the datum. If the pressure at section (1) is 165 kPa, find the pressure at section (2). 10
- 14.** Find the maximum power that can be transmitted through a 2 km long pipe and 80 cm diameter which is supplying water from a dam to power station. The supply head is 90 m of water. Take friction factor, $f = 0.008$. 10
- 15.** Water flows through a pipe of 40 cm diameter and 120 m long with a velocity of 3 m/sec. Find the head lost due to friction by using Chezy's formula. Take Chezy's constant as 55. 10
- 16.** Explain the schematic layout of a hydraulic power with a neat sketch. 5+5
- 17.** Explain the working principle of centrifugal pump with a diagram. 5+5
- 18.** Explain the components of a pneumatic system with a schematic diagram. 5+5

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